

第一次project记录

Part I. Some definitions

Definition 1 (*solution*) :

Given a two dimensional autonomous system :

$$\begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

where

$$\begin{cases} \dot{x} := \frac{dx}{dt} \\ \dot{y} := \frac{dy}{dt} \end{cases}$$

We define :

$$\begin{cases} \dot{x} = \phi(t) \\ \dot{y} = \psi(t) \end{cases} \text{ is a solution } \iff \begin{cases} \phi'(t) = P(\phi(t), \psi(t)) \\ \psi'(t) = Q(\phi(t), \psi(t)) \end{cases}$$

Definition 2 (*Singular point*) :

Given a vector field $v(x) := (P(x), Q(x))$

We say that x_* is a *singular point* of v if $v(x_*) = 0$

Remark :

If x_* is a *Singular point* , then $X = x_*$ is a constant solution of the system.

Definition 3 (*Phase curve*) :

Phase curve $:= \{X : \exists t \text{ s.t } X = (\phi(t), \psi(t))\}$

Remark :

1. The image of a solution is a phase curve.

2. The phase curve is a closed curve $\iff \phi, \psi$ are periodic

Definition 4 (*Limit cycles*) :

A *limit cycle* is an isolated closed curve in the phase plane of a two dimension autonomous system.

Isolated means that neighboring curve are not closed, they spiral either toward or away from the closed curve

Example :

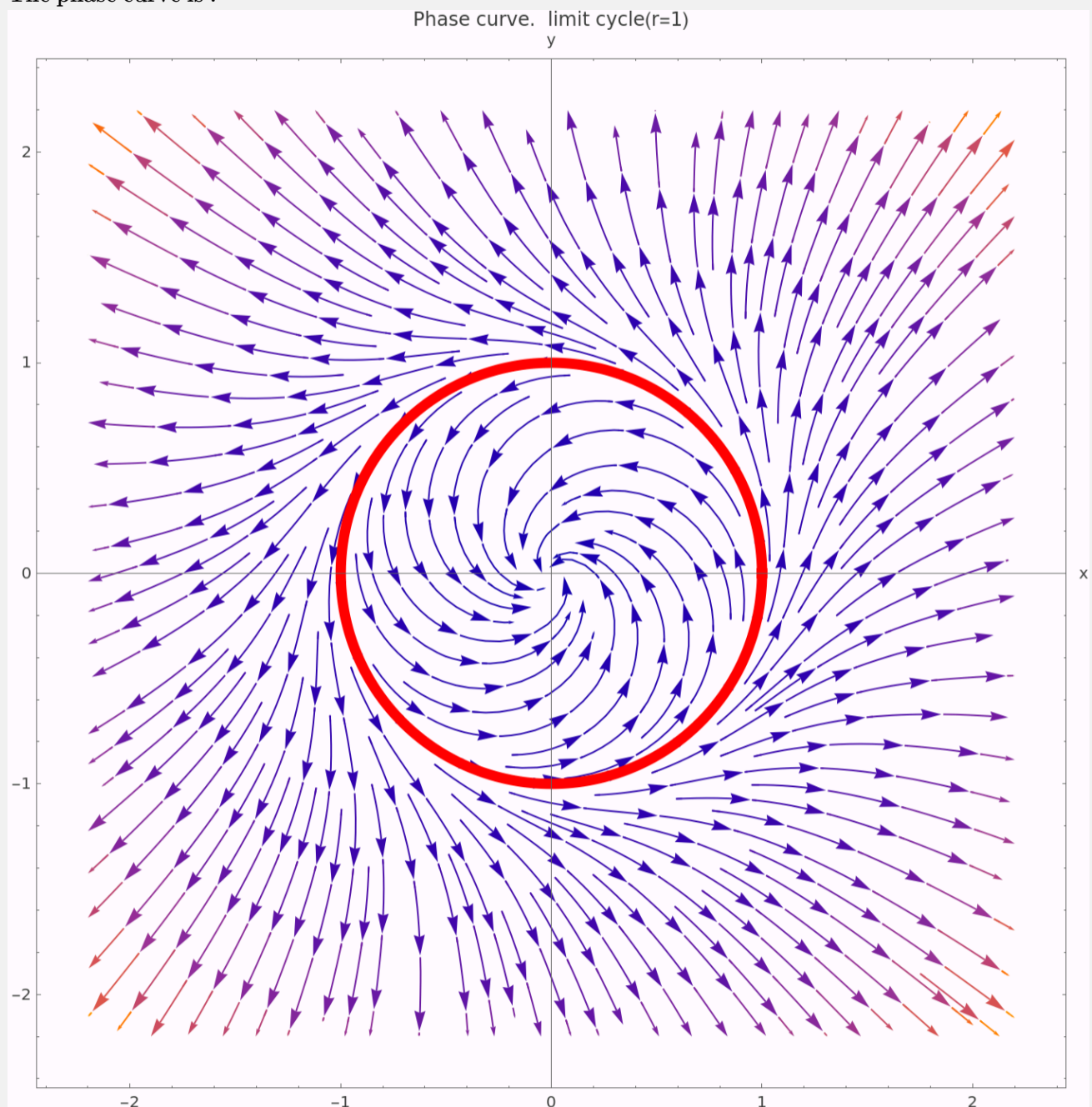
Consider the following system :

$$\begin{cases} \dot{x} = -y + x(x^2 + y^2 - 1) \\ \dot{y} = x + y(x^2 + y^2 - 1) \end{cases}$$

The system can be written as :

$$\begin{cases} \dot{r} = r(r^2 - 1) \\ \dot{\theta} = 1 \end{cases}$$

The phase curve is :



Famous problem :

Given :

$$\begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

1. (16th Hilbert problem) is $\#(\text{limit cycle}) < \infty$?
2. for $\deg(P), \deg(Q) = 2$. What is the max $\#$?

Definition 5 (First integral) :

$I(x, y)$ is a *first integral* of the system if :

$$I(\phi(t), \psi(t)) = \text{constant for all solutions}$$

Remark :

$$I : \mathbb{R}^2 \rightarrow \mathbb{R}, I \in C^1(\mathbb{R}^2)$$

$$I \text{ is a First Integral} \iff \frac{\partial I}{\partial x} P + \frac{\partial I}{\partial y} Q \equiv 0$$

Definition 6 (Complex form of system) :

$$\dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

where

$$A, B, C \in \mathbb{C}$$

is the complex form of two dimensional autonomous system.

Definition 7 (*Hamiltonian system*) :

We say system :

$$\begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

is a *Hamiltonian system* if :

$$\exists H : \mathbb{R}^2 \rightarrow \mathbb{R} \text{ s.t } \begin{cases} \dot{x} = -H_y \\ \dot{y} = H_x \end{cases}$$

Remark :

H is a first integral.

Definition 8 (*Symmetric system*) :

We say system :

$$\begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

is *Symmetric* if :

$$\begin{cases} (\text{P is even}): P(-x, y) = P(x, y) \\ (\text{Q is odd}): Q(-x, y) = -Q(x, y) \end{cases}$$

Part II. The problems

• Question 1

Consider the two dimensional autonomous system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

Suppose our domain is simply connected , Is the following statment correct?

$$\text{div}V = 0 \iff A \text{ is a Hamiltonian system}$$

• Question 2

Suppose we have the two dimensional autonomous system :

$$A : \begin{cases} \dot{x} = P(x, y) \\ \dot{y} = Q(x, y) \end{cases}$$

And the complex form of A is given by:

$$\tilde{A} : \dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z}$$

Supoose :

$$A := A_1 + iA_2$$

$$\begin{aligned} B &:= B_1 + iB_2 \\ C &:= C_1 + iC_2 \\ \mu &:= (A_1, A_2, B_1, B_2, C_1, C_2) \end{aligned}$$

Find the condition of μ such that $\tilde{A}$ is a *Hamiltonian system*.